

Annio: A Web-based Tool for Annotating Medical Images with Ontologies

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Abstract—We describe a web-based, volumetric image annotation tool that is based entirely on HTML5/CSS3 web presentation technologies. The annotation tool can be used on a wide variety of volumetric medical image formats. The application interfaces with ontology web services so that the annotations are labeled with well-defined terms.

Keywords—medical imaging; ontologies; annotation

I. INTRODUCTION

Annotating images is an important step in increasing the computational usefulness of images and to facilitate integration with other data. Recent advances in web technologies (e.g. the HTML5 canvas) enables purely web-based imaging applications. We describe Annio, an ontology-based image annotator that runs entirely within a modern, standards-compliant web browser. Annio allows a user to interrogate a volumetric image, and annotate points, circles, and polygons with each annotation labeled using an ontology. Annio has been implemented as a plugin within an instance of a Midas repository, allowing for in-place annotation with the data never leaving the repository.

II. METHODS

After interpolation to an isotropic voxel size using linear interpolation, the image is converted to three series of orthogonal PNG images that can be efficiently transmitted and displayed in a standard web browser. Conversion is based on SimpleITK which can process a wide variety of image formats including. The 8-bit channels of a PNG image are used to transmit the full (typically 16-bit) data of the original image by splitting bytes into the channels of the PNG image. Annio allows full adjustment of the window and level values of the grayscale (or rainbow) mapping, panning and zooming of the image, and interrogating the voxel values while browsing in either a 2D or 3D mode.

Annotation labels are obtained by matching typed free text to terms in an ontology. As the user types in free text, the top 40 matches to the text from the ontology are returned, with the list being continually updated as the user types. We are currently accessing the RadLex ontology hosted at the National Center for Biomedical Ontologies. The ontology name, the ontology term, and the concept identifier (e.g. "RID545") are stored in an XML scheme along with the world coordinate of each point in the annotation and the unique identifier for the image in the repository.

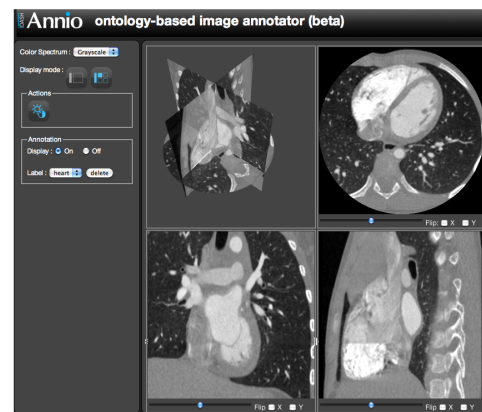


Figure 1: The 3D browsing interface.

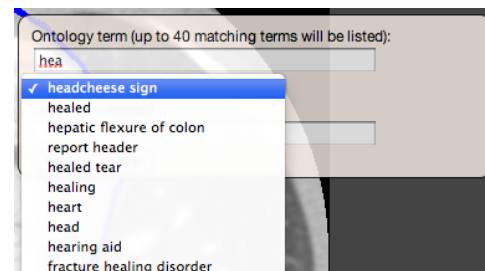


Figure 2: The ontology term selection interface.

III. RESULTS

Figure 1 shows the 3D interface to Annio while Fig. 2 shows the dynamic ontology selection. Annio has been used to annotate pulmonary and cardiac images. Feedback from collaborating radiologists is being used to enhance usability.

IV. CONCLUSION

Annio is a fully web-based image annotator that can sit on top of an image repository for convenient image annotation without the user having to download data or locally install any software. We are currently improving Annio's imaging functionality while also extending data types to be annotated to include time-series and text.

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